Figure6B

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Intro

Functional diversity loss simulated by increasing water temperature scenarios and comparing semesters to reproduce the results of Figure 6B from the original publication. Seasonal dynamics of the coastal microbiome and its association with environmental factors.

1. Set the environment

```
library(tidyverse)
library(vegan)
source("../scripts/resources/increasing_temp_simulation.R")
```

2. Load data

`asvs_workable.tsv` contains the rarefied ASV abundance profiles, with samples as rows and ASVs as columns.
`samo_metadata_workable.tsv` contains the metadata for the samples. `date2season2community.tsv` is a table mapping the date, season, and community columns.

```
ABUND <- read_tsv("../data/opus_workable.tsv.gz", show_col_types = FALSE)
METADATA <- read_tsv("../data/samo_metadata_workable.tsv", show_col_types = FALSE)
DATE2SEASON2COMMUITY <- read_tsv("../data/date2season2community.tsv", show_col_types = FALSE)
```

3. Convert ABUND to long format and map metadata

```
ABUND_long <- ABUND %>%
              left_join(x = .,
                        y = METADATA %>% dplyr::select(Date, Temperature_C),
                        by = c("Date")) %>%
              filter(abundance > 0) %>%
              left_join(x = .,
                        y  = DATE2SEASON2COMMUITY,
                        by = "Date")
```

4. Compute temperature range for each semester

```
semesters2range <- ABUND_long %>%
                        group_by(Community) %>%
                        summarise(min_temp = min(Temperature_C),
                                  max_temp = max(Temperature_C))
```

```r
s12range <- semesters2range %>% filter(Community == "S1")
s22range <- semesters2range %>% filter(Community == "S2")
```

5. Compute temperature range for each OPU

```r
opus2range <- ABUND_long %>%
              group_by(opu_id) %>%
              summarise(min_temp = min(Temperature_C),
                        max_temp = max(Temperature_C)) %>%
              rename(seq_id = opu_id)
```

6. Simulate diversity loss in semester S1

```r
ABUND_wide <- ABUND %>%
              pivot_wider(names_from = opu_id,
                          values_from = abundance,
                          values_fill = 0) %>%
              column_to_rownames(var = "Date")

s1_i <- DATE2SEASON2COMMUITY %>%
        filter(as.character(Date) %in% rownames(ABUND_wide)) %>%
        filter(Community == "S1") %>%
        pull(Date) %>%
        as.character()

ABUND_s1 <- ABUND_wide[s1_i,] %>% as.matrix()

s2_i <- DATE2SEASON2COMMUITY %>%
  filter(as.character(Date) %in% rownames(ABUND_wide)) %>%
  filter(Community == "S2") %>%
  pull(Date) %>%
  as.character()

ABUND_s2 <- ABUND_wide[s2_i,] %>% as.matrix()
```

7. Simulate diversity loss in semester S1

```r
temp_increase <- seq(0, 3, 0.2)

s1_temp_sim <- increasing_temp_simulation(abund_wide = ABUND_s1,
                                          abund_wide_ref = ABUND_s2,
                                          seqs2range = opus2range,
                                          semester2range = s12range,
                                          temp_increase = temp_increase)
```

8. Simulate diversity loss in semester S2

```r
s2_temp_sim <- increasing_temp_simulation(abund_wide = ABUND_s2,
                                          abund_wide_ref = ABUND_s1,
                                          seqs2range = opus2range,
                                          semester2range = s22range,
                                          temp_increase = temp_increase)
```

9. Extract diversity values from nested lists

```
s1_temp_div_df <- map_dfr(s1_temp_sim, ~ {
  tibble(
    temp = .x$temp,
    diversity = .x$div$diversity,
    rich_obs = .x$div$rich_obs,
    rich_chao1 = .x$div$rich_chao1,
    semester = "S1"
  )
})

s2_temp_div_df <- map_dfr(s2_temp_sim, ~ {
  tibble(
    temp = .x$temp,
    diversity = .x$div$diversity,
    rich_obs = .x$div$rich_obs,
    rich_chao1 = .x$div$rich_chao1,
    semester = "S2"
  )
})

temp_div_long_df <- rbind(s1_temp_div_df, s2_temp_div_df) %>%
  pivot_longer(cols = c("diversity", "rich_obs", "rich_chao1"),
               names_to = "metric", values_to = "value") %>%
  group_by(temp, semester, metric) %>%
  summarize(mean = mean(value),
            sd = sd(value))
```

```
## `summarise()` has grouped output by 'temp', 'semester'. You can override using the `.groups` argumen
```

```
temp_div_long_df
```

```
## # A tibble: 96 x 5
## # Groups:   temp, semester [32]
##     temp semester metric      mean       sd
##    <dbl> <chr>    <chr>      <dbl>    <dbl>
##  1   0   S1       diversity  100   0
##  2   0   S1       rich_chao1 100   7.11e-15
##  3   0   S1       rich_obs   100   0
##  4   0   S2       diversity  100   0
##  5   0   S2       rich_chao1 100   7.11e-15
##  6   0   S2       rich_obs   100   0
##  7   0.2 S1       diversity   99.1 2.81e+ 0
##  8   0.2 S1       rich_chao1  93.6 1.93e+ 1
##  9   0.2 S1       rich_obs    93.6 1.92e+ 1
## 10   0.2 S2       diversity   99.6 1.40e+ 0
## # i 86 more rows
```

10. Identify significant differences in diversity loss

```
temp_div_long_df2 <- rbind(s1_temp_div_df, s2_temp_div_df) %>%
  pivot_longer(cols = c("diversity", "rich_obs", "rich_chao1"),
               names_to = "metric", values_to = "value")
```

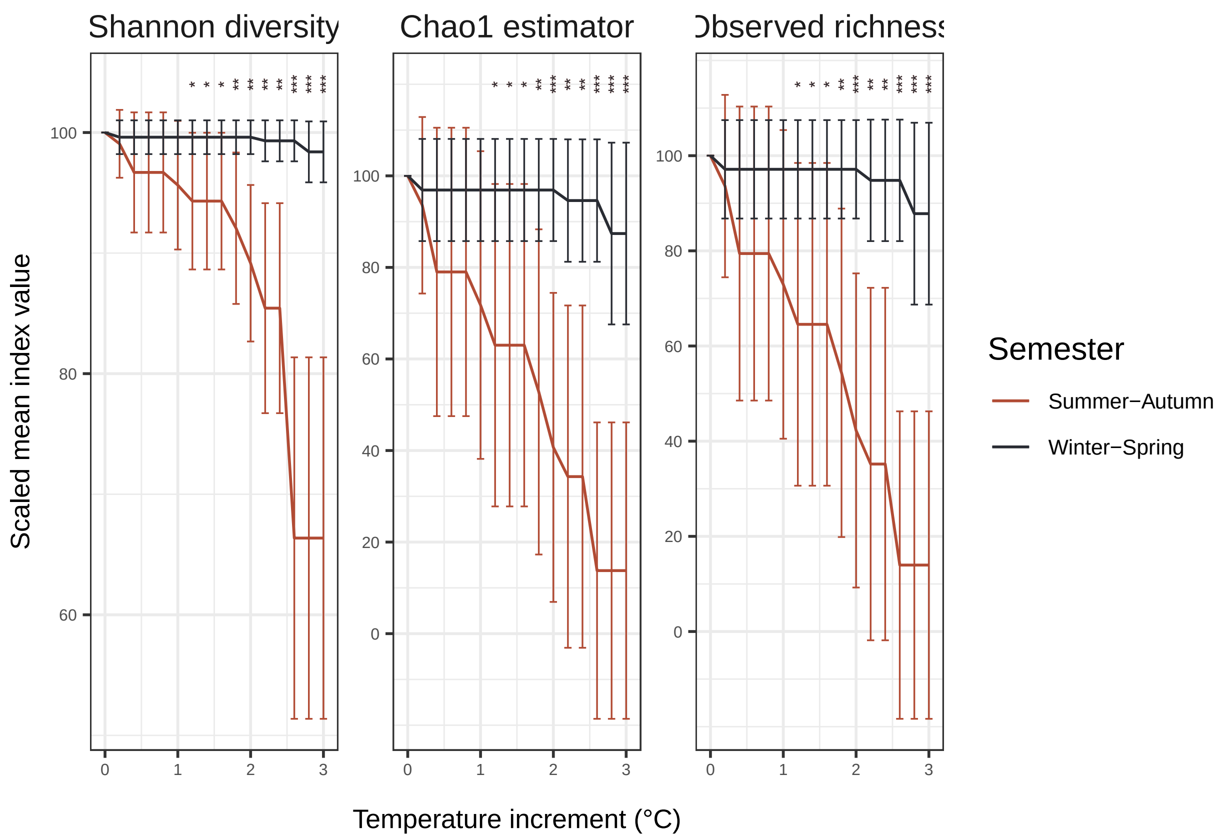
```r
format_pvalues <- function(anova_welch) {

  p_value <- anova_welch$p.value

  if (is.na(anova_welch$p.value) == T) {

    return(NA)
  } else {

    if (p_value < 0.001) {
      p_value <- "<0.001"
    } else {
      if (p_value < 0.01) {
        p_value <- "<0.01"
      } else {
        if (p_value < 0.05) {
          p_value <- "<0.05"
        } else {
          p_value <- ">0.05"
        }
      }
    }
  }
  return(p_value)

}

for (i in temp_increase[-1]) {

  temp_div_df_i <- temp_div_long_df2 %>%
    filter(temp == i)


  # diversity
  welch_anova_i_div <- temp_div_df_i %>%
    filter(metric == "diversity") %>%
    oneway.test(value ~ semester,
                data = .,
                var.equal = FALSE)

  rows_i <- temp_div_long_df$temp == i & temp_div_long_df$metric == "diversity"
  temp_div_long_df[rows_i, "p_value"] <- welch_anova_i_div$p.value
  temp_div_long_df[rows_i, "p_value_formatted"] <- format_pvalues(welch_anova_i_div)

  # richness observed
  welch_anova_i_rich_obs <- temp_div_df_i %>%
    filter(metric == "rich_obs") %>%
    oneway.test(value ~ semester,
                data = .,
                var.equal = FALSE)

  rows_i <- temp_div_long_df$temp == i & temp_div_long_df$metric == "rich_obs"
  temp_div_long_df[rows_i, "p_value"] <- welch_anova_i_rich_obs$p.value
```

```
  temp_div_long_df[rows_i, "p_value_formatted"] <- format_pvalues(welch_anova_i_rich_obs)

  # richness Chao
  welch_anova_i_rich_chao1 <- temp_div_df_i %>%
    filter(metric == "rich_chao1") %>%
    oneway.test(value ~ semester,
                data = .,
                var.equal = FALSE)

  rows_i <- temp_div_long_df$temp == i & temp_div_long_df$metric == "rich_chao1"
  temp_div_long_df[rows_i, "p_value"] <- welch_anova_i_rich_chao1$p.value
  temp_div_long_df[rows_i, "p_value_formatted"] <- format_pvalues(welch_anova_i_rich_chao1)

}
```

11. Plot diversity loss

```
text_size <- 8

plot_labels <- c("diversity" = "Shannon diversity",
                 "rich_obs" = "Observed richness",
                 "rich_chao1" = "Chao1 estimator")

temp_div_long_df <- temp_div_long_df %>%
                    mutate(y_max = if_else(metric == "diversity", 104, NA),
                           y_max = if_else(metric == "rich_obs", 115, y_max),
                           y_max = if_else(metric == "rich_chao1", 120, y_max))

div_plot <- ggplot(temp_div_long_df, aes(x = temp, y = mean, color = semester)) +
            geom_line() +
            facet_wrap(~metric, scales = "free_y", labeller = as_labeller(plot_labels)) +
            theme_minimal() +
            xlab("Temperature increment (°C)") +
            ylab("Scaled mean index value") +
            scale_color_manual(values = c("#b14b34","#292c33"),
                               labels = c("S1" = "Summer-Autumn", "S2" = "Winter-Spring"),
                               name = "Semester") +
            geom_errorbar(aes(ymin = mean - sd, ymax = mean + sd),  linewidth = 0.3,  width = 0.1) +
            scale_y_continuous(breaks = seq(0, 100, 20)) +
            theme_bw() +
            theme(
              axis.text.x = element_text(size = text_size-2),
              axis.text.y = element_text(size = text_size-2),
              axis.title.x = element_text(size = text_size +2, margin = unit(c(4,0,0,0), "mm")),
              axis.title.y = element_text(size = text_size +2, margin = unit(c(0,2,0,0), "mm")),
              strip.text = element_text(size = text_size+4),
              strip.background = element_blank(),
              # increase legend text
              legend.text = element_text(size = text_size),
              legend.title = element_text(size = text_size+4)
            )  +
            geom_text(data = temp_div_long_df %>% filter(p_value_formatted == "<0.001"),
                      aes(x = temp, y = y_max, label = "***", vjust = 0.7),
```

```
                    size = 2, angle = 90, show.legend  = FALSE) +
          geom_text(data = temp_div_long_df %>% filter(p_value_formatted == "<0.01"),
                    aes(x = temp, y = y_max, label = "**", vjust = 0.7),
                    size = 2, angle = 90, show.legend  = FALSE) +
          geom_text(data = temp_div_long_df %>% filter(p_value_formatted == "<0.05"),
                    aes(x = temp, y = y_max, label = "*", vjust = 0.7),
                    size = 2, angle = 90, show.legend  = FALSE)

div_plot
```



12. Print session info

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 20.04.6 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.9.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.9.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C               LC_TIME=en_US.UTF-8        LC_COLLATE=en_
##  [6] LC_MESSAGES=en_US.UTF-8    LC_PAPER=en_US.UTF-8       LC_NAME=C                  LC_ADDRESS=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
```

```
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] fANCOVA_0.6-1     maps_3.4.2        corrgram_1.14     ggpubr_0.6.0      ggcorrplot_0.1.4.1 v
##  [8] permute_0.9-7     lubridate_1.9.3   forcats_1.0.0     stringr_1.5.1     dplyr_1.1.4        p
## [15] tidyr_1.3.1       tibble_3.2.1      ggplot2_3.5.1     tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.5      xfun_0.48         rstatix_0.7.2     tzdb_0.4.0        vctrs_0.6.5       tools_
##  [9] fansi_1.0.6       highr_0.11        cluster_2.1.6     pkgconfig_2.0.3   Matrix_1.7-0      lifec
## [17] tinytex_0.53      munsell_0.5.1     carData_3.0-5     htmltools_0.5.8.1 yaml_2.3.10       Formul
## [25] car_3.1-3         MASS_7.3-61       abind_1.4-8       nlme_3.1-166      tidyselect_1.2.1  diges
## [33] splines_4.4.2     cowplot_1.1.3     fastmap_1.2.0     colorspace_2.1-1  cli_3.6.3         magri
## [41] withr_3.0.1       scales_1.3.0      backports_1.5.0   bit64_4.5.2       timechange_0.3.0  rmark
## [49] hms_1.1.3         evaluate_1.0.1    knitr_1.48        mgcv_1.9-1        rlang_1.1.4       glue_
## [57] R6_2.5.1
```